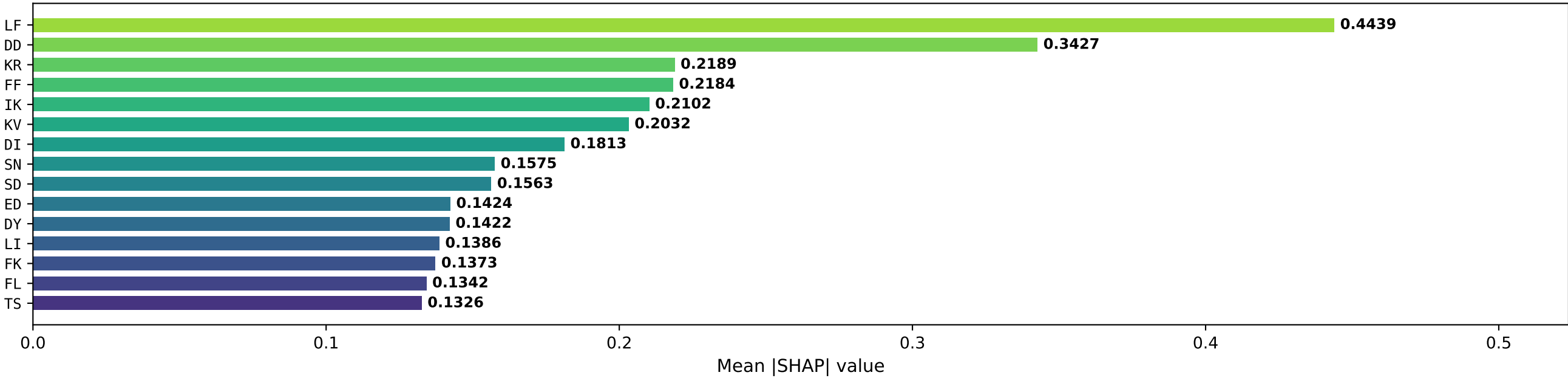
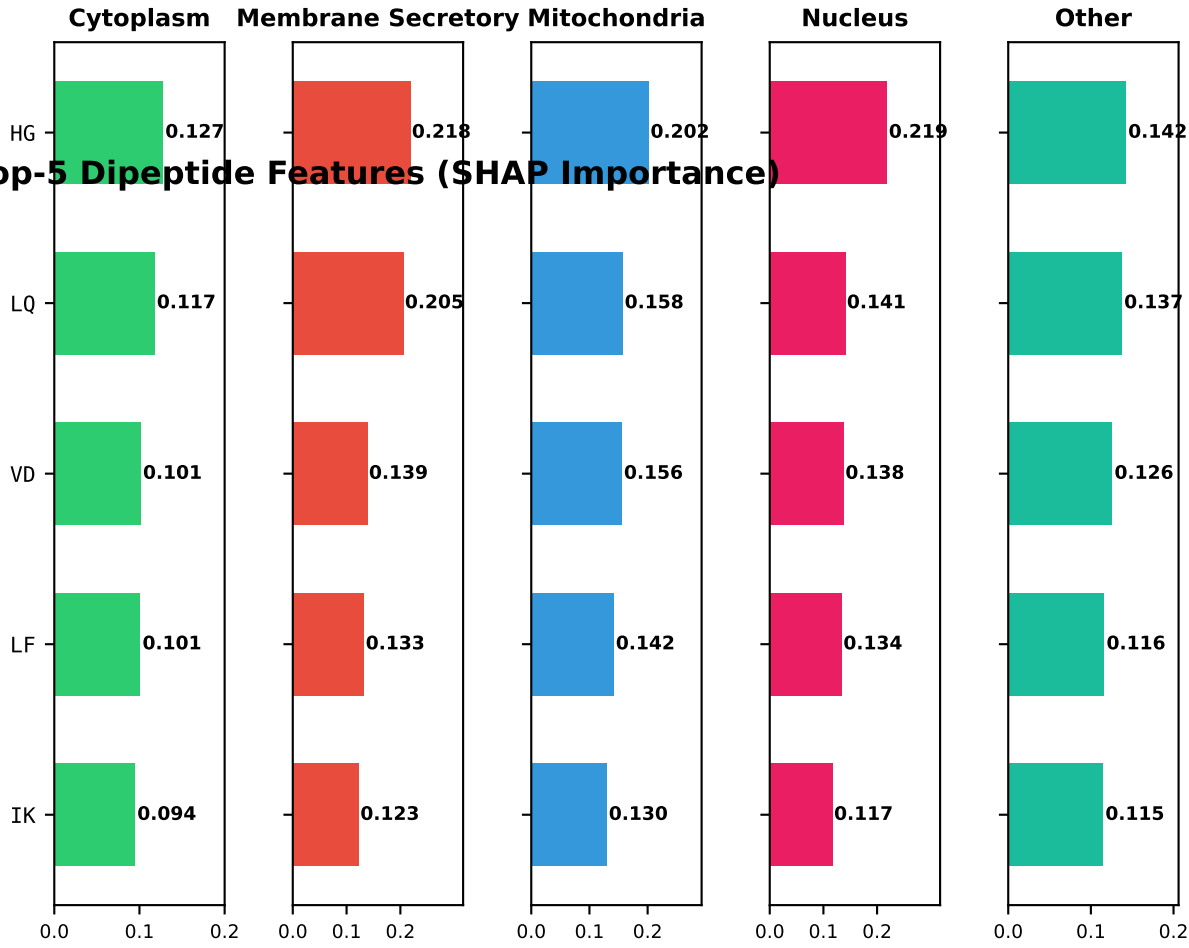


A Overall Top-15 Dipeptide Features (SHAP Importance)



B Per-Class Top-5 Dipeptide Features (SHAP Importance)



C Biological Corroboration

Dipeptide	Class	Ratio	p-adj	Verdict
FF	Membrane	2.69	<0.001	Confirmed (signal peptide)
LF	Membrane	1.86	<0.001	Confirmed (signal peptide)
FI	Membrane	1.76	<0.001	Novel finding
KR	Nucleus	1.41	<0.001	Confirmed (NLS)
RK	Nucleus	1.27	0.002	Partly supported
DD	Nucleus	1.60	<0.001	Enriched (acidic patch)
DD	Mitochondria	0.62	<0.001	Depleted (matches MTP)
KG	Mitochondria	1.45	<0.001	Novel finding
EE	Mitochondria	0.75	<0.001	Depleted (matches MTP)
HG	Cytoplasm	1.12	0.04	Novel finding

Confirmed (lit.) Novel / Enriched Depleted / Partial